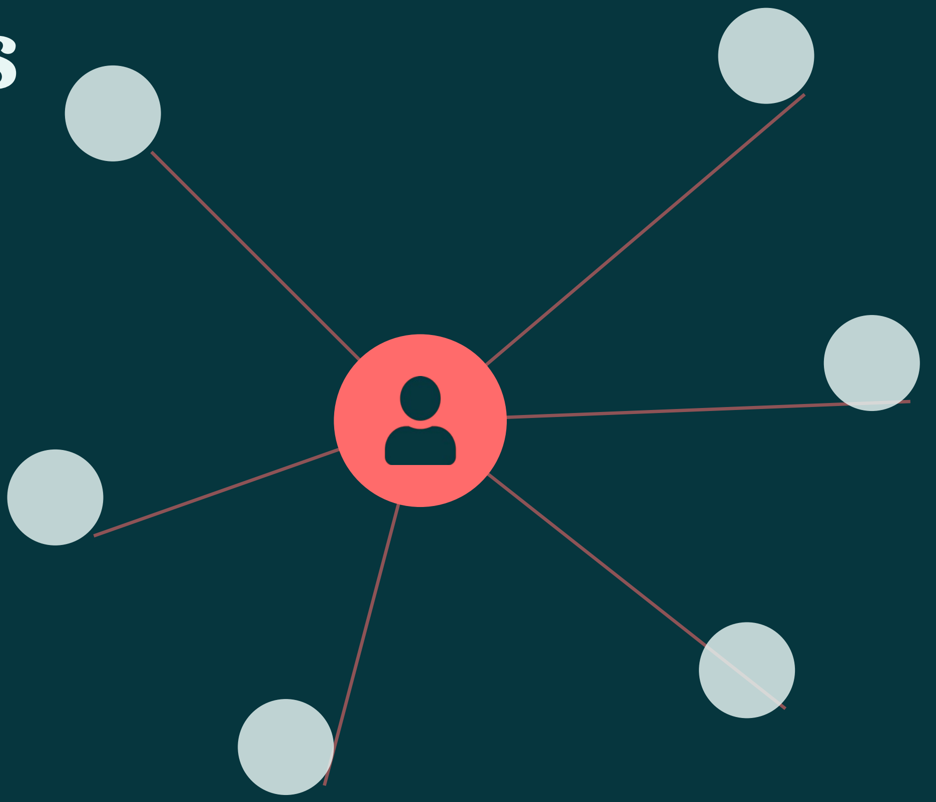
E-COMMERCE · A BEGINNER'S GUIDE

How Recommendation Systems Work

A hands-on, jargon-light introduction

For aspiring data scientists



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WHAT YOU'LL LEARN

Five things you'll be able to explain by the end

The problem	Why recommendations exist and what they solve
Content-based filtering	Recommending by item features
Collaborative filtering	Recommending from behavior patterns
Evaluation	How we measure if recommendations are good
Getting started	Tools and datasets to build your first model

THE PROBLEM, INTUITIVELY

Imagine a store with a billion shelves

You can't browse it all

Nobody can scroll a billion products. Most never get seen.

Too many choices freeze us

Faced with endless options, people often buy nothing.

A good guide helps

A recommender is like a clerk who remembers everyone's taste.

Why it pays off

17%

of global online orders driven by AI recommendations & personalization

+7%

higher average order value for retailers using generative AI

Salesforce Shopping Index, 2024.

THE CORE IDEA

Predict a score, then sort by score

Almost every recommender does the same thing: give each item a “how much will this user like it?” score, then show the highest-scoring items first.

Input

What we know: your past clicks/buys, item details, the moment.

Model

A formula that outputs a relevance score for each item.

Ranked output

Items sorted from highest score to lowest — top ones shown.

APPROACH 1 — CONTENT-BASED FILTERING

“You liked this, so here's something similar”

Recommend items whose features match what the user already liked.

Example

A shopper liked: *“Trail Running Shoe”*

Tags: sport · outdoor · lightweight · rubber sole

So we look for items sharing those tags:

- ✓ **Hiking Sandals** outdoor · lightweight
- ✓ **Gym Trainers** sport · rubber sole
- ✗ **Office Loafer** formal · leather

Good to know

- \+ Works for brand-new items (we know their features)
- \+ Easy to explain: “because it's also outdoor gear”
- − Can feel repetitive — stays close to what you know
- − Needs good item data (tags, descriptions)

APPROACH 2 — COLLABORATIVE FILTERING

“People like you also liked…”

Ignore item features. Learn from behavior: find users with similar taste, then recommend what they liked.

	Item A	Item B	Item C	Item D
You	★★★★	★★	—	?
Ana	★★★★	★★	—	★★★★
Ben	★	★★★★	★★★	★
Cara	★★★★	★★	★	★★★★

Reading the table

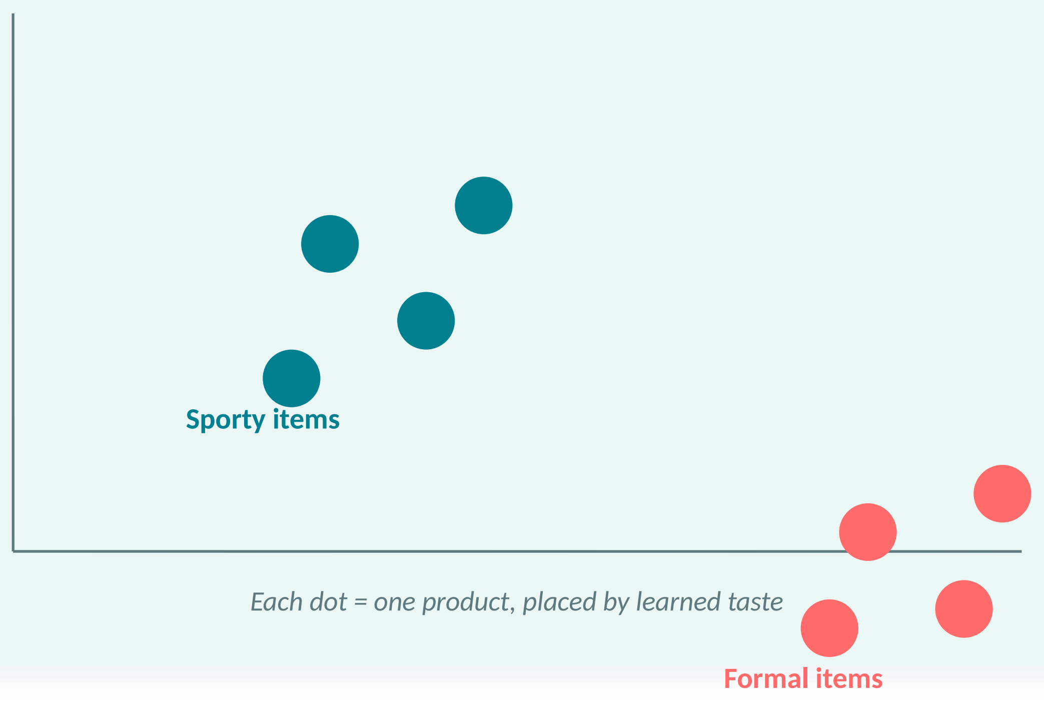
1. You and Ana rate Items A & B almost identically.
2. Ana loves Item D (★★★★). You haven't tried it.
3. So we predict you'll like Item D too → recommend it.
4. That's “user-based” CF. “Item-based” flips it: find items rated alike.

Turn taste into coordinates; nearby = similar

Real rating tables are huge and mostly empty (most people rate almost nothing).

Embeddings compress each user and item into a short list of numbers — a point in space.

Items close together are similar; we recommend items near what you like.



Each dot = one product, placed by learned taste

THE DATA BEHIND IT

What users say vs. what they do

Explicit signals

Users tell us directly

- Star ratings you give
- Reviews you write
- Likes & wishlists
- Preferences you pick

Implicit signals

We watch behavior

- Clicks and taps
- Things you buy
- How long you look
- Items added to cart

Explicit data is clear but rare (most people never rate). Implicit data is everywhere but noisy — a click isn't always a “like.”

THE COLD-START PROBLEM

What to do when there's no history yet

New user

Someone just signed up — we don't know their taste yet.

New item

A product nobody has interacted with — nothing to learn from.

Common fixes

- Show popular items to everyone at first
- Ask a quick “what are you into?” at sign-up
- Recommend new items by their features (content-based)
- Use clues: location, device, time of day

Two metrics, explained with a tiny example

We recommended 3 items. Did the user like them?

Item A	*liked*
Item B	*liked*
Item C	*ignored*

Precision@3 = 2 of 3 = 0.67

NDCG also rewards putting the liked items higher up the list.

Two kinds of testing

Offline (lab)

Test on past data. Cheap and fast. Metrics: precision@k, recall, NDCG.

Online (real users)

Run an A/B test on live traffic. Metrics: click-through, conversion.

REAL SYSTEMS IN THE WILD

The same ideas, at enormous scale

Amazon

Item-based collaborative filtering — the “bought this also bought” idea you just learned.

Taobao / Alibaba

Graph embeddings place a billion items in “taste space” to power the “Guess What You Like” feed.

JD.com (Jingdong)

Deep-learning models match shoppers to products across a massive catalog in milliseconds.

BEGINNER PITFALLS & GLOSSARY

Mistakes to dodge, terms to know

Common pitfalls

- Trusting offline metrics only — always A/B test
- Data leakage: testing on data the model already saw
- Recommending only popular items (popularity bias)
- Ignoring cold start for new users and items

Quick glossary

CF — Collaborative Filtering

AOV — Average Order Value

CTR — Click-Through Rate

NDCG — ranking quality, top-weighted

Embedding — item/user as a vector

YOUR NEXT STEPS

Build your first recommender this week

Get a dataset

Start with MovieLens — small, clean, and classic for learning.

Pick a library

Try Surprise, LightFM, or implicit for ready-made models.

Build a baseline

Start with popularity, then item-based collaborative filtering.

Evaluate it

Split train/test, measure precision@k, then iterate.

Tip: get a simple baseline working end-to-end first — then improve one piece at a time.

KEY TAKEAWAYS

What you can now explain

1. Recommenders score every item, then show the highest-scoring ones
2. Content-based = match item features; collaborative = learn from behavior
3. Embeddings turn taste into coordinates; nearby items are similar
4. Implicit data is plentiful but noisy; watch out for cold start
5. Measure offline (precision@k, NDCG) but decide with online A/B tests

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Now you have the map — go build your first recommender!

Thank you

BONUS SECTION

How real systems grew up

A friendly tour of how Amazon, Taobao, and JD.com built their recommenders — from the classics to today's AI — and who benefits most. Every number comes from 2024–2026 sources.

Amazon — recommendation system history

How Amazon's recommendations grew from “also bought” lists into a chat assistant.

PRE-GPT ERA

GPT / LLM ERA

1998	**Item-to-item CF**	The classic “people who bought this also bought.”
2003	**Published (IEEE)**	Amazon explained the method in a research paper.
2017	**Whole-store personalization**	Whole pages start to get personalized.
2024	**Rufus + COSMO**	Rufus: a chat assistant that answers shopping questions.
2025	**Rufus at scale**	Rufus remembers you and tailors its replies.

GPT-era impact

~$12B	300M+	+60%
extra sales in 2025 (Amazon)	people use Rufus	more likely to buy (engaged users)

Sources: Amazon Science (2024); Amazon Q4 2025 earnings; COSMO — ACM SIGMOD 2024; Linden et al., IEEE Internet Computing (2003, historical).

BONUS · HOW REAL SYSTEMS GREW UP

Taobao / Alibaba — recommendation system history

How Taobao went from "taste vectors" to AI that reasons about what you want.

PRE-GPT ERA		GPT / LLM ERA		
2018	2018	2025	2025	2025
EGES graph embeddings	**Deep Interest Network**	**LUM — Large User Model**	**URM universal retriever**	**RecGPT**
Turns browsing into "taste" vectors for the feed.	Learns what you're interested in right now.	A giant "user model" that scales like an LLM.	One AI model that fetches items for any goal.	An assistant that reasons about what you want.

GPT-era impact

+2.9%

more clicks in Taobao search test (2025)

18.8% → 46.2%

more new-category discovery (2025)

Sources: Wang et al., KDD 2018 (EGES); Zhou et al., KDD 2018 (DIN); Yan et al., 2025 (LUM); Jiang et al., 2025 (URM, arXiv:2502.03041); Yi et al., 2025 (RecGPT).

BONUS · HOW REAL SYSTEMS GREW UP

JD.com (Jingdong) — recommendation system history

How JD.com moved from early AI recommenders to generative AI in production.

PRE-GPT ERA

GPT / LLM ERA

2017

Telepath

An early AI recommender that also "sees" images.

2021

Semantic retrieval + re-rank

Smarter search that understands meaning, not just words.

2025

GenCTR

AI that predicts clicks for search ads.

2025

GRAM

AI that fetches the right products (generative retrieval).

2025

CFRAG

Blends AI understanding into the results it recalls.

GPT-era impact

Generative

AI now predicts clicks in production (2025)

Generative

AI now fetches products for search (2025)

Sources: Zhao et al. (Telepath), 2017; Li et al., 2021; Kong et al., 2025 (GenCTR, arXiv:2507.11246); JD GRAM & CFRAG, 2025.

Who does AI help most? (by shopper type)

Different shoppers get different benefits. The big idea: AI helps new users most, because it can read words — not just past clicks.

New shoppers

We barely know them yet

- **Why it helps:** AI reads product text, so it can suggest even with no history
- **Alibaba:** shows far more new categories: 18.8% → 46.2% (2025)
- **Netflix:** +8% better at surfacing brand-new titles (2025)

Occasional shoppers

They drop in now and then

- **Amazon Rufus:** answers their questions in plain language (2025)
- **Taobao RecGPT:** guesses intent instead of just past clicks (2025)
- **Plain-language search:** type what you mean, not exact keywords

Loyal shoppers

Lots of history to learn from

- **Amazon:** engaged shoppers buy 60% more often (2025)
- **Alibaba:** big "user models" turn long history into +2.9% clicks (2025)
- **Takeaway:** more history = sharper personalization

Sources (2025): Jiang et al. (URM); Yan et al. (LUM); Amazon Q4 2025; Li et al., Nov 2025 (Netflix); cold-start video study, Aug 2025. Platform-level per-segment figures are rarely disclosed; cross-industry studies used as directional evidence.

APPENDIX · SOURCES

Where these facts come from (2024–2026)

Amazon — Rufus	Amazon Science, “The technology behind Amazon's GenAI shopping assistant, Rufus” (2024); scale & sales via Amazon Q4 2025 earnings.	*amazon.science/blog/...-rufus*
Amazon — COSMO	Commonsense knowledge graph built with LLMs, ACM SIGMOD 2024.	*amazon.science (COSMO, SIGMOD 2024)*
Alibaba — URM	Jiang et al., “LLM as Universal Retriever in Industrial-Scale Recommender System” (2025).	*arxiv.org/abs/2502.03041*
Alibaba — LUM & RecGPT	Yan et al., “Large User Model” (2025); Yi et al., “RecGPT Technical Report” (2025).	*arXiv 2025 (LUM); RecGPT report, Aug 2025*
JD.com — GenCTR & GRAM	Kong et al., “Generative CTR Prediction...” (2025); JD “GRAM: Generative Retrieval and Alignment” (2025).	*arxiv.org/abs/2507.11246*
User-class / cold-start	Netflix cold-start (Li et al., Nov 2025); video-platform cold-start study (Aug 2025); cold-start LLM survey (2025).	*arXiv 2025*

Note: pre-GPT milestones (item-to-item CF 1998/2003; EGES & DIN 2018; JD Telepath 2017) are historical; all quantitative effectiveness figures above are from 2024–2026 publications.